

Revision — parallelism and perpendicularity in space

The whole year of solid geometry, reduced to eight statements and one picture each.

LESSON 63–64

2 × 40 MIN

GEOMETRY 10, CHAPTERS II–IV REVIEW

P1 SPATIAL REVIEW

LESSON CLOCK

15'

25'

25'

20'

15'

Positions	15 min
Parallelism	25 min
Perpendicularity	25 min
Angles	20 min
Homework	15 min

LEARNING OBJECTIVES

- State the positions of two lines, a line and a plane, and two planes.
- Use the criteria for a line parallel to a plane and for two parallel planes.
- Use the criterion for a line perpendicular to a plane, and the three-perpendiculars theorem.
- Compute the angle between a line and a plane and between two planes.

TEXTBOOK

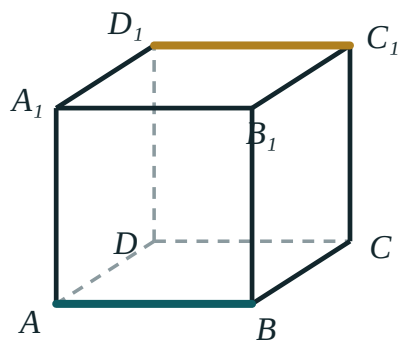
National	pp. 257–266
Cambridge	Pure Mathematics 1, pp. 77–80
Workbook	Revision set G1

01

Explanation

Positions

PAIR	POSSIBLE POSITIONS
two lines	intersecting · parallel · skew
a line and a plane	the line lies in it · parallel · meets it in one point
two planes	coincident · parallel · meeting in a line



AB and C_1D_1
never meet and
are not parallel:
they are skew

Skew lines — not parallel, and never meeting. The case that has no analogue in the plane.

SKREW IS THE CASE THAT ONLY SPACE HAS

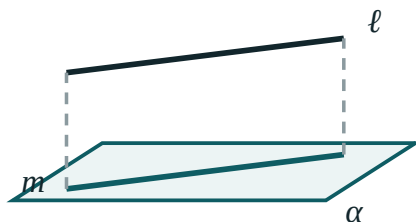
Two lines that do not meet need not be parallel — they may simply lie in different planes. In a cube the twelve edges give 18 parallel pairs, 24 intersecting pairs and 24 skew pairs, and $18 + 24 + 24 = 66 = C(12, 2)$.

Parallelism

THE TWO CRITERIA

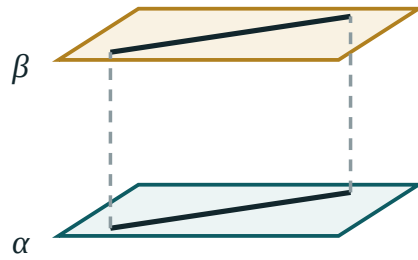
- A line is **parallel to a plane** if it is parallel to some line lying in that plane.
- Two planes are **parallel** if two intersecting lines of one are parallel to two intersecting lines of the other.

In both, the word **intersecting** is essential: two parallel lines of one plane parallel to two of the other prove nothing.



$l \parallel m$ and m lies in α , so $l \parallel \alpha$

A line parallel to a plane — it is parallel to a line drawn in the plane.



$\alpha \parallel \beta$ — a third plane cuts them in parallel lines

Two intersecting lines of each, in parallel pairs — that is the criterion.

Two consequences used constantly:

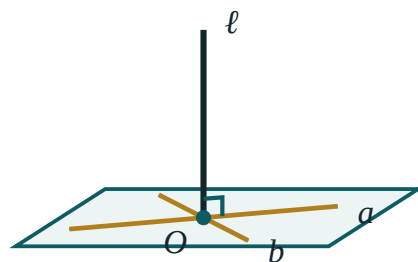
- If $\alpha \parallel \beta$ and a third plane cuts both, the two lines of intersection are parallel.
- Segments of parallel lines between parallel planes are equal.

Perpendicularity

THE CRITERION, AND THE THEOREM

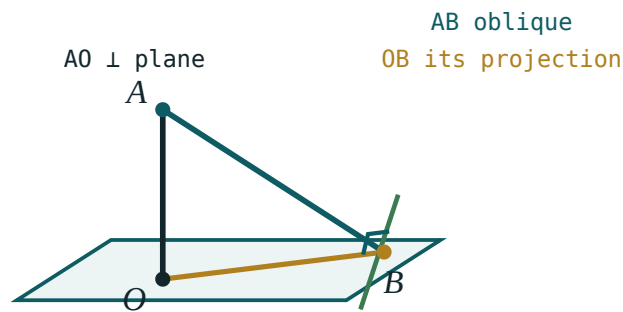
A line is **perpendicular to a plane** if it is perpendicular to two **intersecting** lines of that plane. It is then perpendicular to **every** line of the plane.

Three perpendiculars. A line in a plane is perpendicular to an oblique if and only if it is perpendicular to the oblique's projection.



$\ell \perp a$ and $\ell \perp b$, with a and b intersecting at O therefore ℓ is perpendicular to the whole plane

Perpendicular to two intersecting lines — and therefore to all of them.



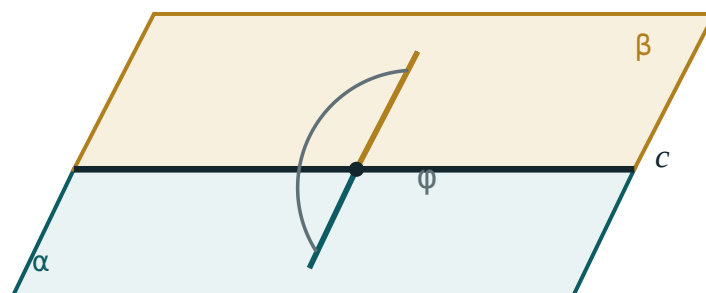
a line of the plane through B is $\perp$ AB
exactly when it is $\perp$ OB

*The oblique, its projection, and the line in the plane —
perpendicular to one iff to the other.*

The three-perpendiculars theorem is what turns a space problem into a plane one: it lets you drop from a slanted segment to its shadow and work entirely in the base.

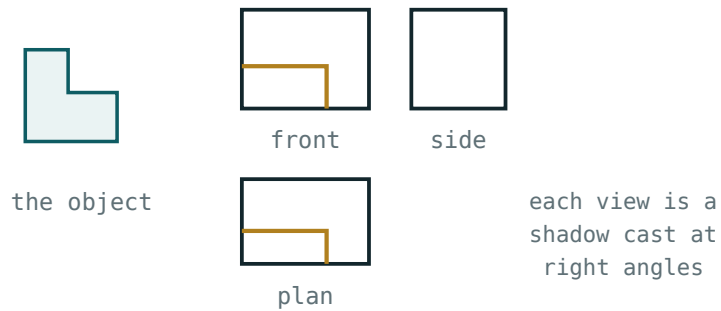
Angles

ANGLE	HOW TO CONSTRUCT IT
line and plane	between the line and its orthogonal projection
two planes (dihedral)	between two rays perpendicular to the edge, one in each face
two skew lines	translate one until they meet; the angle between the translates



both arms are perpendicular to the edge c
the angle φ between them is the dihedral angle

*The linear angle of a dihedral — both rays
perpendicular to the edge.*



three orthogonal projections determine the solid

The projection is the shadow cast straight down; the angle is measured to it.

EVERY ANGLE QUESTION BECOMES ONE RIGHT TRIANGLE

Find the foot of the perpendicular, name the projection, and a right triangle appears with the slant as hypotenuse. Then $\cos \theta = \frac{\text{projection}}{\text{slant}}$ or $\tan \theta = \frac{\text{height}}{\text{projection}}$ — plane trigonometry from that point on.

02 Worked examples

EXAMPLE 1 In a cube $ABCD A_1 B_1 C_1 D_1$ of edge a , find the angle between AC_1 and the base.

① The projection of AC_1 on the base is AC .

② $AC = a\sqrt{2}$, $CC_1 = a$

③ $\tan \theta = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}}$

④ $\theta \approx 35.3^\circ$

ANSWER

$\approx 35.3^\circ$

EXAMPLE 2 In the same cube, find the angle between the planes ABC_1D_1 and the base.

- 1 The edge of the dihedral is AB .
- 2 $BC \perp AB$ in the base; $BC_1 \perp AB$ in the upper face.
Both perpendicular to the edge.
- 3 $\tan \theta = \frac{CC_1}{BC} = 1$
- 4 $\theta = 45^\circ$

ANSWER

45°

EXAMPLE 3 Prove that a line perpendicular to two intersecting lines of a plane is perpendicular to a third line of that plane.

- 1 Let $l \perp a$ and $l \perp b$, with $a \cap b = O$.
- 2 Any line c of the plane through O is a combination of directions of a and b .
- 3 So $l \perp c$ as well.
This is the criterion, and its point.

ANSWER

Perpendicular to two intersecting lines means perpendicular to all

EXAMPLE**4**

A point is 12 cm from a plane. Two obliques of lengths 13 and 15 cm are drawn. Find their projections.

$$① \quad \sqrt{169 - 144} = 5$$

$$② \quad \sqrt{225 - 144} = 9$$

③ Projections 5 cm and 9 cm.

The longer oblique has the longer projection.

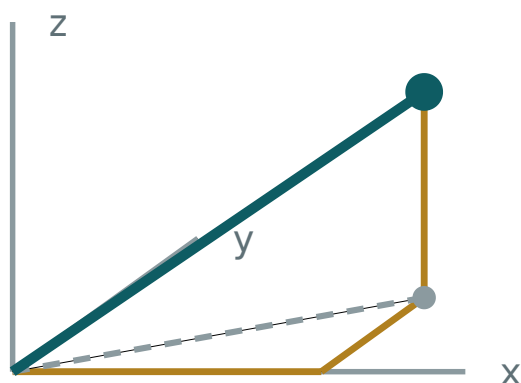
ANSWER

5 cm and 9 cm

03 Interactive model

Use a wire cube and a torch: the shadow on the desk is the orthogonal projection, and every angle question becomes visible.

Lines and planes in a box



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Skew lines	Ayqash to'g'ri chiziqlar	Скрещивающиеся прямые
Parallel to a plane	Tekislikka parallel	Параллельная плоскости
Parallel planes	Parallel tekisliklar	Параллельные плоскости
Perpendicular to a plane	Tekislikka perpendikulyar	Перпендикулярная плоскости
Three perpendiculars theorem	Uch perpendikulyar teoremasi	Теорема о трёх перпендикулярах
Angle between a line and a plane	To'g'ri chiziq va tekislik orasidagi burchak	Угол между прямой и плоскостью
Dihedral angle	Ikki yoqli burchak	Двугранный угол
Orthogonal projection	Ortogonal proyeksiya	Ортогональная проекция
Oblique	Og'ma	Наклонная
Foot of the perpendicular	Perpendikulyar asosi	Основание перпендикуляра

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Two lines that never meet are:

always parallel

parallel or skew

always skew

intersecting

A line is perpendicular to a plane if it is perpendicular to:

one line of it

two intersecting lines of it

two parallel lines of it

the edge

The angle between a line and a plane is measured to:

any line of the plane

its orthogonal projection

the normal

the edge

A dihedral angle is measured by:

any two rays

two rays perpendicular to the edge

the edge itself

the projection

The three-perpendiculars theorem connects:

two planes

an oblique and its projection

two skew lines

two obliques

In a cube, the number of skew edge pairs is:

18

24

12

66

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Three positions of two lines in space

ANSWER Intersecting, parallel, skew

2. Three positions of a line and a plane

ANSWER In it, parallel, meeting in a point

3. A line $\perp$ to two intersecting lines of a plane is:

ANSWER Perpendicular to the plane

4. The angle between a line and a plane is measured to:

ANSWER Its projection

5. Space diagonal of a cube of edge a

ANSWER $a\sqrt{3}$

6. Face diagonal of a cube of edge a

ANSWER $a\sqrt{2}$

7. Projection of an oblique 13 from a point 12 from the plane

ANSWER 5

Medium · the standard question

7 PROBLEMS

1. Angle between AC_1 and the base in a cube

ANSWER $\approx 35.3^\circ$

2. Angle between ABC_1D_1 and the base in a cube

ANSWER 45°

3. Angle between two space diagonals of a cube

ANSWER $\approx 70.5^\circ$ or $\approx 109.5^\circ$

4. Number of parallel edge pairs in a cube

ANSWER 18

5. Number of intersecting edge pairs in a cube

ANSWER 24

6. Number of skew edge pairs in a cube

ANSWER 24

7. Two obliques of 15 and 20 from a point 12 from a plane: projections

ANSWER 9 and 16

Hard · stretch and reason

7 PROBLEMS

1. Distance from a vertex of a cube of edge a to the diagonal of the opposite face

ANSWER $\frac{a\sqrt{6}}{3}$

2. Angle between AB_1 and BC_1 in a cube

ANSWER 60°

3. Distance between two skew edges of a cube of edge a

ANSWER a

4. A rectangular box $3 \times 4 \times 12$: the angle of the space diagonal with the base

ANSWER $\approx 67.4^\circ$

5. Prove that if $\alpha \parallel \beta$ and γ cuts both, the lines of intersection are parallel

ANSWER They lie in γ and cannot meet

6. A point 6 from a plane: the locus of feet of obliques of length 10

ANSWER A circle of radius 8

7. A regular tetrahedron of edge a : the angle between a lateral edge and the base

ANSWER $\approx 54.7^\circ$

07 Homework

Homework — 5 tasks

Draw the cube for every question; the marks live in the picture.

- List the three positions of two lines, of a line and a plane, and of two planes, with one sketch each.
- In a cube of edge 6 cm, find the angle between the space diagonal and the base.
- In the same cube, find the dihedral angle between the plane through A, B, C_1 and the base.
- A point is 8 cm from a plane; an oblique from it is 17 cm long. Find the projection.

5. State the three-perpendiculars theorem and draw the picture that goes with it.